

Day 4: Tasks

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).



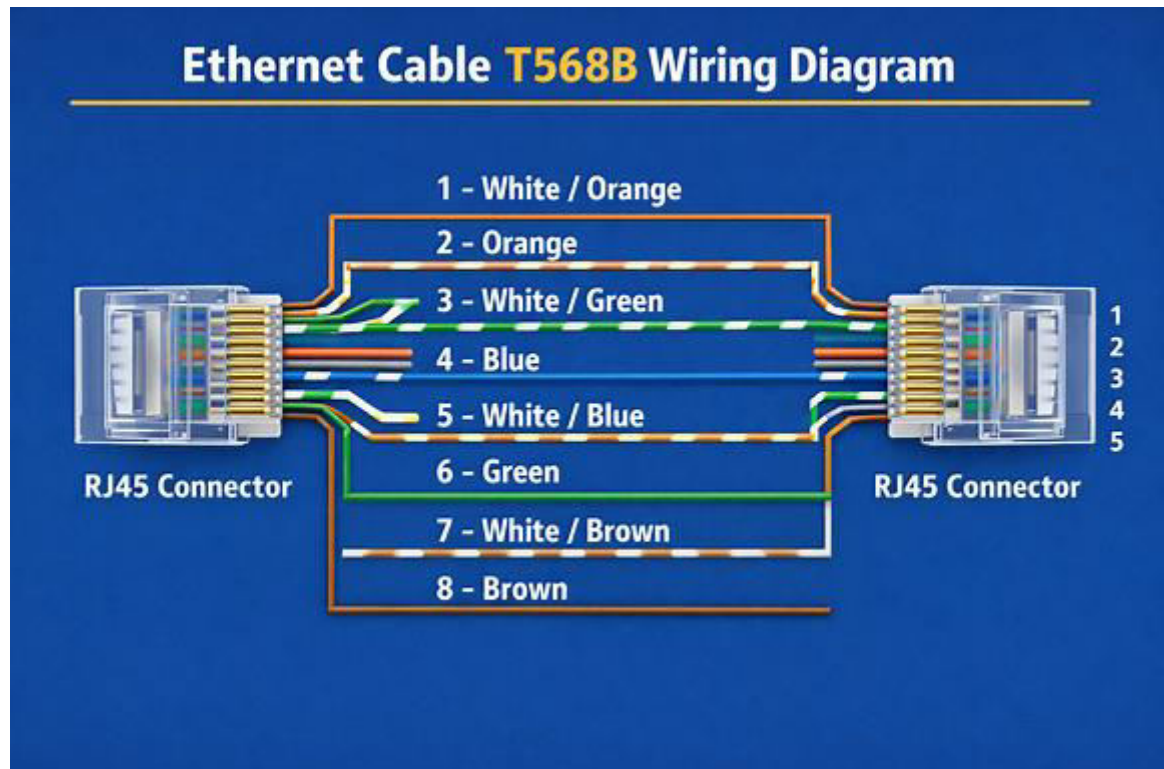
1. Three Key Differences Between Ethernet Cables and Fiber Optic Cables

Feature	Ethernet Cable	Fiber Optic Cable
Data Transmission	Uses electrical signals through copper wires.	Uses light signals through glass or plastic fibers.
Speed & Bandwidth	Good for normal home and office networks.	Much faster and supports very high bandwidth.
Distance	Works best for shorter distances (up to about 100 meters).	Can carry data over several kilometers without major signal loss.

Real-World Examples :

- **Ethernet Cable Example:**
 - Used to connect a home WiFi router to a desktop computer, smart TV, or gaming console.
 - It provides a stable and reliable internet connection inside homes and small offices.
- **Fiber Optic Cable Example:**
 - Used as the backbone network in large places such as college campuses, data centers, and internet service provider (ISP) networks.
 - It is preferred because it can transfer large amounts of data at very high speeds over long distances.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.



Step-by-Step Guide to Attach an RJ45 Connector to a CAT6 Cable

Tools Required:

- CAT6 Ethernet cable
- RJ45 connector
- RJ45 crimping tool
- Wire stripper or cutter
- Cable tester (optional but recommended)

Steps

Step 1: Cut the Cable

- Cut the CAT6 cable to the required length using a cable cutter.

Step 2: Strip the Outer

Jacket

- Remove about 2–3 cm of the outer insulation using the wire stripper.
- Be careful not to damage the wires inside.

Step 3: Untwist the Wires

- You will see 4 twisted pairs (8 wires).
- Untwist them gently and straighten each wire with your fingers.

Step 4: Arrange the Wires (T568B Standard)

Arrange the wires in this exact order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Step 5: Trim the Wires

- Hold the wires tightly together.
- Cut all 8 wires evenly so they are the same length (about 12–13 mm exposed).

Step 6: Insert the Wires into the RJ45 Connector

- Slide the wires carefully into the RJ45 connector.
- Make sure each wire goes into the correct slot.
- Push the cable in until the outer jacket also enters the connector for a secure grip.

Step 7: Crimp the Connector

- Insert the RJ45 connector into the crimping tool.
- Squeeze the handles firmly until the connector is locked onto the cable. This presses the metal pins into the wires and secures the connector.

Step 8: Test the Cable

- Use a cable tester to check that all 8 pins are connected correctly.
- If the test passes, the cable is ready to use.

- 4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.**



Cable Type: Ethernet (CAT6)

Tool Used: RJ45 Crimper

Reasons:

1. Ethernet CAT6 cables provide high speed and low latency, which are important for smooth LAN gaming.
2. The distance between PCs and the network switch in a gaming café is usually short, so fiber optic cables are not required.
3. Ethernet cables and RJ45 connectors are more affordable, helping keep the setup within budget.
4. An RJ45 crimper is used to attach RJ45 connectors to the CAT6 cables, making installation quick and easy.